Backpack LiDAR-Based SLAM With Multiple Ground Constraints for Multistory Indoor Mapping

Baoding Zhou, *Member, IEEE*, Haoquan Mo, Shengjun Tang, *Member, IEEE*, Xing Zhang, *Member, IEEE*, and Qingquan Li

***Abstract*— High-quality 3-D point cloud maps are essential for precise indoor environment modeling. However, constructing such maps in multistory indoor environments is challenging due to the presence of narrow nonstructural spaces, such as staircases, corners, and corridors with similar textures. Simultaneous localization and mapping (SLAM) in these scenes is particularly difficult, as cumulative errors can lead to incorrect loop closures and drastic degradation in map quality. To address these challenges, this article proposed an SLAM method based on multiple ground constraints pose optimization (MGCPO), which uses a backpack light detection and ranging (LiDAR) system. The proposed method includes two novel modules. First, a regression analysis-based scenarios recognition (RASR) module provides a reference for the construction of ground constraints. Second, based on different scene detection results, the MGCPO module constrains the sensor pose using the floor plane to reduce localization errors and effectively decrease loop closure detection errors. Qualitative experiments demonstrate that our proposed method outperforms state-of-the-art methods in challenging scenarios. Quantitative experiments show that our method achieves an error rate of just 1.06% using only LiDAR sensors.**

***Index Terms*— Backpack light detection and ranging (LiDAR) system (BLS), ground constraints, multistory indoor environments, pose graph optimization, simultaneous localization and mapping (SLAM).**

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Baoding Zhou and Haoquan Mo are with the Institute of Urban Smart Transportation and Safety Maintenance and the Key Laboratory for Resilient Infrastructures of Coastal Cities, Ministry of Education, Shenzhen University, Shenzhen 518060, China, and also with the College of Civil and Transportation Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: bdzhou@szu.edu.cn; 2110474149@email.szu.edu.cn).

Shengjun Tang and Xing Zhang are with the School of Architecture and Urban Planning, Shenzhen University, Shenzhen 518060, China, and also with the State Key Laboratory of Subtropical Building, South China University of Technology, Guangzhou 511442, China (e-mail: shengjuntang@szu.edu.cn; xzhang@szu.edu.cn).

Qingquan Li is with the Guangdong Key Laboratory of Urban Informatics and the Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen University, Shenzhen 518060, China (e-mail: liqq@szu.edu.cn).

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I. Introduction

HIGH-QUALITY point cloud maps are usually used to solve the positioning and navigation issues of mobile robots and autonomous vehicles. In addition, point cloud map plays an important role in spatial model reconstruction for digital cities. Among the techniques used for point cloud data collection, light detection and ranging (LiDAR) systems have evolved various mapping platforms, such as terrestrial laser scanners (TLSs) and mobile laser scanners (MLSs) [1], [2]. While the scanning accuracy of MLS is lower than that of TLS, MLS is widely adopted in applications due to its efficiency, especially in indoor mapping scenarios. Common types of MLS include vehicle LiDAR systems, backpack LiDAR systems (BLS), and handheld LiDAR systems. For indoor environments, staircase scenarios included in the scanning route will limit the passing of wheeled vehicle scanners. Handheld scanners rely on expensive multibeam LiDAR or complex rotary mechanical structures to obtain dense point cloud maps. BLS is rarely limited by the scanning scenarios and can be easily developed into a multisensorial system, making it a hot topic in research and application for indoor mapping. Gong et al. [3] used two laser scanners to assemble their dual-LiDAR BLS, completing mapping in global navigation satellite system (GNSS)-denied environments, such as indoor multistory buildings and underground parking lots. Wen et al. [4] used a BLS to achieve indoor mapping and integrated backpack indoor data with TLS outdoor data. Karam et al. [5], [6] customized a BLS for indoor mapping, which combines 2-D laser scanners to generate 3-D point clouds.

Plane-based registration methods have been widely applied in mapping tasks. Pavan and dos Santos [7] segmented the plane from dense points to solve the transformation between TLS scanning stations and refine the registration with global closed-form constraints. Dos Santos et al. [8] used an image-based segmentation technique for detecting diverse plane regions in the RGB images. The associated 3-D centroid and the correspondent disparity values are used to refine the initial transformation calculated by feature-based methods. Both these works proved that plane-segmentation-based methods improve the accuracy of points cloud registration. However, dense points containing sufficient areas of the plane are required for plane segmentation, sparse points collected with BLS might not contain such clouds, especially in small-scale scenes, such as long corridors, narrow staircases, and

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multistory indoor environments. Though the required cloud could be gained with scan accumulation, map-to-map registration is more suitable for the implementation of the backend, rather than LiDAR odometry (LO) on the BLS simultaneous localization and mapping (SLAM) system [31]. With the proposal of ground constraints, the prior normal vector of the ground plane offers a solution to reduce the localization drift during SLAM in poor-geometric feature scenes, such as multistory indoor environments. Lightweight and ground-optimized LOAM on variable terrain (LeGO-LOAM) [9] adopted a local ground constraints strategy. The extracted ground points from a single LiDAR were aligned with the ground points within consecutive previous scans to estimate the 3-DOF of the 6-DOF pose of an unmanned ground vehicle (UGV). Su et al. [10] introduced a depth factor to calculate the roughness to eliminate the outliers of ground points based on LeGo-LOAM [9]. This work is also devised for the LiDAR system on UGV. Koide et al. [11] used a global ground constraints strategy which assumed the existence of a unique ground and completed indoor SLAM with a BLS equipped with a Velodyne-HDL32E LiDAR. However, only one LiDAR captures very limited ground points in a narrow indoor environment, causing a descending performance of this method. Chen et al. [12] customized a fast ground segmentation method for their dual-LiDAR BLS but still assumed that there was only one global ground in the environment. However, this assumption does not adjust to a multistory indoor environment containing a staircase and several ground planes.

With the development over the past decades, SLAM has evolved into feature-based and probability-based methods. Served as the representative of feature-based methods, the LO and mapping (LOAM) [13] determines the point sequence and ring parameters by calculating yaw and pitch angles. Then edge and surface points in each ring are extracted by calculating the curvature of points. Finally, the ego-motion is estimated by jointly optimizing point-to-edge and point-to-surface residual using the Levenberg–Marquardt method. Both feature extraction and pose estimate modules in LOAM [13] assume that the interval between laser beams is almost consistent, which does not hold in some multi-LiDAR systems, such as dual-LiDAR BLS. Jiao et al. [14] proposed a robust odometry and mapping for multi-LiDAR systems with online extrinsic calibration, which solved this problem to some extent but was more suited to vehicle multi-LiDAR systems. Using all points streamed from LiDAR, probability-based methods are able to be compatible with multi-LiDAR systems. In terms of probability-based methods, iterative closest points (ICPs) [15], [16] and normal distribution transformation (NDT) [17] are classical methods and have been improved by researchers since their publishments. In addition, frame-to-submap registration has been adopted in most SLAM systems since its proposal in Cartographer [18]. These registration algorithms usually maintain a local map with a temporal or spatial sliding window strategy, which has been proven to reduce the drift. Generally, the rest cumulative drift will be solved by loop closure correction, which uses descriptor-based methods [19], [20], [21] and direct methods [15], [16], [17] to recognize loop points. Descriptor-based methods discretize a full 3-D point cloud into sectors using polar coordinates. Height or intensity information from each sector is then extracted and encoded into a 2-D matrix to search the loop candidates. Direct methods calculate the fitness score according to the alignment result of NDT [17] or ICP [15], [16], requiring a set of frame candidates provided by the Euclidean distance. All loop match determinations in these methods rely on the texture similarity. However, multistory environments might be composed of diverse floors with similar textures, which might also generate similar descriptors. Therefore, wrong loop matches and corrections are prone to occur in multistory indoor scenes. One of the solutions to reject the wrong loop matching is the limitation of the search range. However, LO could not offer an accurate initial pose for loop candidates' search due to accumulative drift caused by the complex motion of BLS and the nonstructural environment. To overcome this problem, we propose a scene recognition-based pose optimization strategy to gain a low-drift initial pose for loop detection.

Generally, scene recognition methods can be categorized into deep learning methods and nondeep learning methods. Served as the representative of deep learning methods, PointNet [22], [23] series have shown fantastic performance in object detection and scene segmentation. Nevertheless, in our scenario recognition task, the scenario recognition module uses sparse points from the current scan to classify the flat ground and staircase. The results of this simple binary classification will be fused into the pose estimation to improve the SLAM system. Deep learning methods usually require a lot of training and are less explainable. And the deployment of the inference model in SLAM systems is an additional implementation. LOAM [13] detects the degraded environment by calculating the eigenvalues of the approximate Hessian matrix. Bao et al. [24] transformed the point cloud into a depth image and extracted the regression features to identify the monotonous scene. These two methods utilize scenario recognition information to improve the accuracy of horizontal pose estimation. Inspired by those nondeep learning methods, a scenario recognition method is developed to improve the SLAM performance in multistory environments in this article.

Three main contributions regarding the mapping problem in the multistory indoor environments are achieved as follows.

1) This article proposes a regression analysis-based method for dynamic scene classification, which accurately distinguishes two different types of passing scenarios: walking on stairs and walking on a flat surface. The scene classification results are then applied as constraints for localization and loop closure detection in LiDAR-based SLAM processes.
2) Based on multiground geometric constraint, a pose optimization strategy for backpack LiDAR-based SLAM is proposed. The algorithm constructs plane constraints based on different passing scenarios, effectively reducing vertical trajectory drift during the mapping process with backpack-mounted laser scanners (BLSs) and avoiding incorrect loop closure detection between different floors.
3) In order to evaluate the effectiveness of the proposed method, we constructed a dual-LiDAR backpack mapping system and conducted accuracy assessments based

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Fig. 1. Backpack LiDAR system appearance.

on 3-D data collection experiments in different types of scenes. The comparison was conducted with state-of-the-art LiDAR SLAM methods, and the experimental results demonstrated that the proposed method significantly enhances the robustness and accuracy of LiDAR SLAM, especially in multistory indoor mapping and somewhere with repetitive texture on vertical dimensions.

II. Methodology

A. System Overview

The SLAM algorithm proposed in this study is implemented on a BLS equipped with two Velodyne VLP-16 LiDARs [25]. The deployment of the two laser scanners is illustrated in Fig. 1, with one scanner positioned horizontally and the other mounted nearly vertically. A microcomputer with Jetson TX2 (ARM Cortex-A57) is embedded for data collection. And datasets are logged using the robot operation system (ROS1). The coordinate transformation between two LiDARs is calibrated through extrinsic calibration using plane features [26], [27].

The system is implemented using C++ and executed on a workstation with Ubuntu. The framework of the proposed mapping method is illustrated in Fig. 2. Initially, the laser scans from two LiDARs are streamed into the data preprocessing module and merged into a joint scan. To maintain registration speed, the joint points are downsampled using a voxel filter. Simultaneously, the unfiltered points are received by the scenarios recognition thread.

Within the scenarios recognition module, the statistical feature of extracted floor points will be calculated to determine whether BLS is in a flat ground environment. When the recognition module identifies a flat ground scenario, the ground plane equation is fit by RANSAC [26] to provide a constraint for pose optimization. In the pose optimization thread, the constraints are constructed based on the flat ground and odometry. If the condition of detection interval is met, the optimized pose is used to retrieve historical poses. Then new constraints corresponding to ground and loop closure are added to the pose graph. The global optimization is performed with G2O optimizer [30]. Finally, an accurate pose estimation can be obtained and the map is optimized.

B. Preliminaries

1) Joint Pose Graph Optimization: Pose graph optimization is one of the mainstream backend approaches to optimize the pose generated by odometry. In the pose graph model, optimized or fixed parameters, such as sensor poses of keyframes and landmark information, are represented as nodes, while the constraints between these parameters are represented as edges. To optimize the poses, the errors of the edges, usually, are minimized through constructing and solving the least squares problem. Following [29] and [30], the least squares optimization problem is solved using the Levenberg–Marquardt method defined as follows:

$$x^* = \arg\min_x \sum_{j=1}^{m} \sum_{k=1}^{n} e_k^j(x_k, z_k)^T \Omega_k^j e_k^j(x_k, z_k) \tag{1}$$

where x_k and $e_k^j(x_k, z_k)$ denote the node and an error function between the parameters x_k and the measurement z_k of constraints type j, respectively. And Ω_k^j denotes the information matrix. x^* denotes the optimized poses of all keyframe nodes.

2) Hesse Form and Ground-Plane Constraints: The Hesse form (HF) consists of the plane normal vector $n = [n_x, n_y, n_z]^r$ and displacement d between the plane and the origin of reference coordinate frame. The 3-DOF pose estimation [roll, pitch, z] can be gained by solving ground plane constraints. HF π is used to describe the plane equation (2) transformation of ground parameters. The process can be described to that align the ground plane $\pi^k = [n^k, d^k]^T$ in frame f^k to the target ground plane $\pi^j = [n^j, d^j]^T$ in frame f^j using a transformation matrix $T_j^k = [R_j^k | t_j^k]$

$$\begin{cases} n^k = \left(R_j^k\right)^T n^j \\ d^k = d^j - \left(n^j\right) t_j^k. \end{cases} \tag{2}$$

To solve the pose transformation T_j^k, we construct the error e^G for the least-square optimization problem. Inspired by Ma et al.'s [32] work and hdl_graph_slam [11], the error is modeled to $\tau(\pi) = \tau(\varphi, \theta, d)$, where φ, θ, d represent the azimuth angle, the elevation angle, and the intercept in the plane equation, respectively. Then, the corresponding edge in the pose graph model is defined as follows:

$$\tau(\pi) = \left[\arctan\left(\frac{n_y}{n_x}\right), \arctan\left(\frac{n_z}{|n|}\right), d\right] \tag{3}$$

$$e^G\left(x_k, \pi^j\right) = \tau\left(\pi^k\right) - \tau\left(\pi^j\right). \tag{4}$$

C. Data Preprocessing

Before logging data, a system initialization check is performed to avoid a large timestamp interval between the first scan from two LiDARs. Additionally, a data temporal synchronization process, implemented with ROS1, is executed first in

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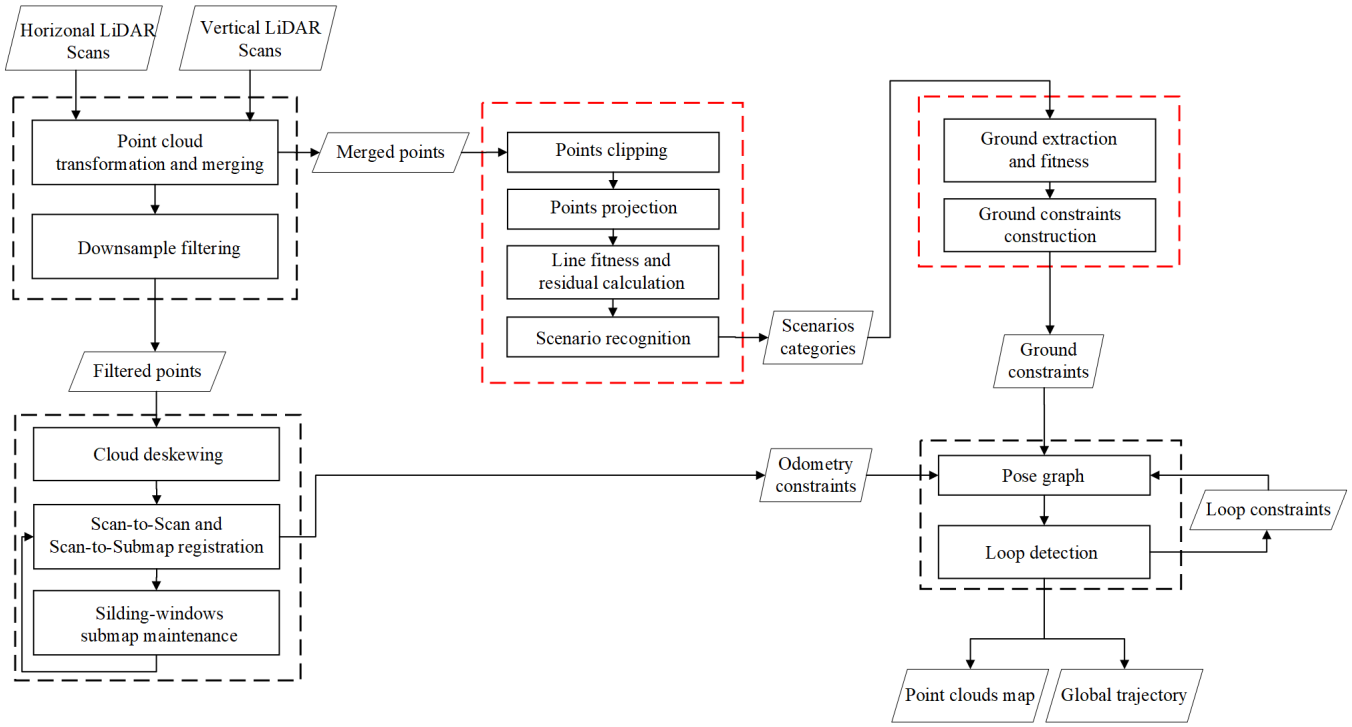


Fig. 2. Proposed algorithm framework. The modules marked with red dashed lines are the main contributions of our work.

the data preprocessing module. This synchronization ensures that the two laser scans with the closest timestamps are merged into a joint scan. After merging, the joint scan is labeled with the earlier timestamp and aligned to the horizontal LiDAR coordinate system using the previously calibrated extrinsic parameters. To maintain registration efficiency, the joint scan is downsampled using a voxel filter with a voxel size of 0.4 m. The distortions of points were eliminated by a uniform-motion model. However, unfiltered points are transmitted to the scenarios recognition module to ensure a sufficient number of floor points for subsequent calculations.

D. LiDAR Odometry

Inspired by Segal et al.'s [33] work and Koide et al.'s [34] work, we introduce Generalized-ICP (G-ICP) [33], a probability-based registration method, to achieve a two-step registration LO for our BLS. As one of the variants of ICP, G-ICP regularizes the eigenvalues of the covariance matrix, making it work as a plane-to-plane ICP. For efficiency, the G-ICP is implemented to a multithread version known as fast generalized iterative closest point (FAST-GICP) [34]. The detailed description of the registration process is shown in the following:

$$\Delta x_i^* = \underset{\Delta x_i}{\operatorname{argmin}} \sum_{j=1}^{n} d_j^T \left(C_j^{f^t} + \Delta x_i C_j^{f^s} \Delta x_i^T \right)^{-1} d_j \tag{5}$$

$$x_0^i = x_{i-1}^i \cdot x_0^{i-1} = \Delta x_i \cdot x_0^{i-1} \tag{6}$$

$$f_o^i = x_0^i * f_b^i \tag{7}$$

$$\begin{cases} M_o^i = M_o^{i-1} + f_o^i, & 0 < i < L_s \\ M_o^i = M_o^{i-1} + f_o^i - f_o^{i-L_s}, & i \geq L_s. \end{cases} \tag{8}$$

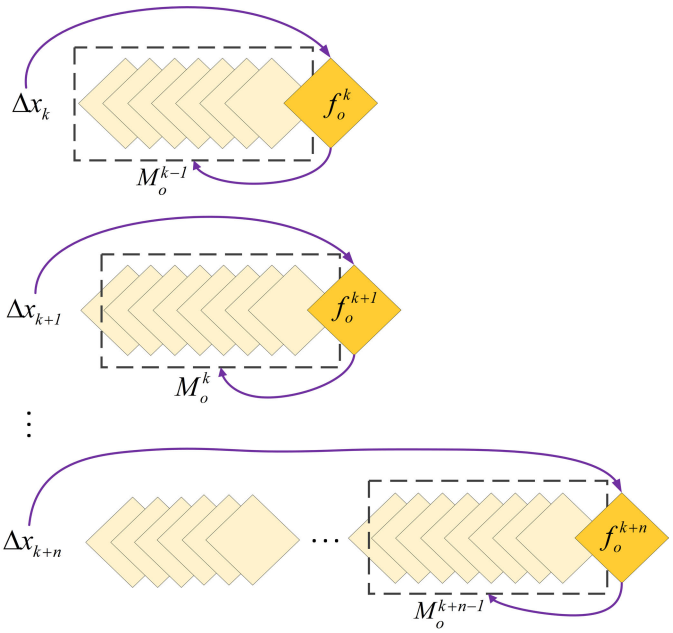


Fig. 3. Submap is maintained by a sliding window using a keyframe strategy.

The current frame was aligned with the previous frame first. Then, the frame was aligned again with the local map comprised of keyframes within sliding window [18], using the initial provided by the frame-to-frame registration. The x_{i-1}^i represents the transformation matrix of the BLS from frame f^i to frame f^{i-1} solved by the twice registration. The G-ICP registration process could be demonstrated in (5). Here, d_j denotes the distance between the corresponding point pairs within source cloud f^s and target cloud f^t. $C_j^{f^t}$ and $C_j^{f^s}$ are

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covariance matrices associated with the corresponding points clusters. The x_0^i in (6) represents the transformation matrix of the BLS from moment i to origin in the odometry coordinate system. The size of the sliding window is set as L_s, which means the submap consistently comprises L_s keyframes after initialization. The injection of keyframes complies with the spatial interval strategy. Converting from Δx_i, if the Δq_w in rotation $[\Delta q_x, \Delta q_y, \Delta q_z, \Delta q_w]$ or displacement Δt calculated by $[\Delta t_x, \Delta t_y, \Delta t_z]$ is larger than the threshold ω or D_t, latest frame f_b^i will be transformed using LO parameter x_0^i and inserted into local map M_o^{i-1} as described in (7) (see Fig. 3). And, from (8), the oldest frame in the window will be discarded. After the procedure earlier, the latest local map M_o^i is assembled.

E. Regression Analysis-Based Scenarios Recognition

In this section, we present a lightweight staircase scene recognition method based on regression analysis. The process begins by extracting a range of floor points from the vertical LiDAR using a clip box. The extracted points are then projected to the plane of the current scan coordinate. Fig. 4 illustrates the detailed steps involved in this process.

The parameters of clip box scale denoted as $[x_{\text{cb}}^a, x_{\text{cb}}^b, y_{\text{cb}}^a, y_{\text{cb}}^b, z_{\text{cb}}^a, z_{\text{cb}}^b]$ are empirically set based on the approximate height of the LiDAR sensor, width of staircase or corridors and scanning scenarios. As depicted in Fig. 4(a), when the operator is walking on flat ground or ascending stairs, we can observe that the height of floor points is distributed within the range of $[-2.9, -1.9]$ m. For scenarios involving descending stairs, the range is adjusted to $[-2.1, -1.4]$ m. Thus, the clip box scales in the z-direction $[z_{\text{cb}}^a, z_{\text{cb}}^b]$ are set as $[-2.9, -1.9]$, $[-2.9, -1.9]$, and $[-2.1, -1.4]$ for three scenarios, walking on flat ground, ascending stairs and descending stairs, respectively. Furthermore, the setting of the clip box scale $[x_{\text{cb}}^a, x_{\text{cb}}^b]$ is used to eliminate points corresponding to the operator's hindrance and is fixed at $[-1.1, -0.4]$ by our experimental experiences.

In Fig. 4(b), the red points labeled by the yellow box represent the extracted floor points. The floor points within the clip box are projected onto the xoy plane to fit an univariate linear equation (9). Then, three regression features gradient k and the intercept b are then determined using the Gaussian–Newton method, which involves solving a least squares problem

$$[k, b]^* = \arg\min_{k,b} \sum_{j=1}^{n} \left| z_j^{f_b^i} - \left(k^* x_j^{f_b^i} + b \right) \right|^2 \tag{9}$$

$$\tilde{z}_j^{f_b^i} = k * x_j^{f_b^i} + b. \tag{10}$$

In statistical analysis, root-mean-square error (RMSE) is commonly used to quantify the dispersion or variability between the predicted values $\tilde{z}_j^{f_l^i}$ calculated by the fit equation and the actual value $z_j^{f_l^i}$. In our case, we use RMSE to calculate the error term as described in the following:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \left(\tilde{z}_j^{f_l^i} - z_j^{f_l^l} \right)^2}. \tag{11}$$

The temporal behavior of the statistical features k, RMSE along with the corresponding recognition results, are depicted in Fig. 5. From Fig. 5, it can be observed that k and RMSE exhibit drastic changes during scenarios transitions. Hence, these features are utilized to differentiate between different scenes. To determine the scene, we employ a threshold value R for RMSE generated by the current scan. If RMSE exceeds R, we further check the value of RMSE to avoid incorrect judgments caused by points corresponding to the operator's feet. Furthermore, if k surpasses a certain threshold K_u, the scenario is identified as walking upward on stairs. For descending stairs scenarios, we consider the condition where k is below K_d. If k falls within a range defined by thresholds $[K_d, K_u]$, it indicates walking on flat ground. The selection of $[K_d, K_u]$ is based on empirical values derived from experiments. To enhance the robustness of regression analysis-based scenarios recognition (RASR), the recognition result for the previous frame is used to determine the current scene when there are insufficient points to fit the line or abnormal results are obtained.

F. Multiple Ground Constraints-Based Pose Optimization Strategy

It is assumed that each floor in the indoor environment consists of a single flat surface, which aligns with the characteristics of most indoor environments. Leveraging this prior information, a reliable map is generated by incorporating multiground constraints into the pose graph optimization process. Fig. 6 demonstrates the multiple ground constraints pose optimization (MGCPO) procedure during descending stairs. And the detailed descriptions are as follows.

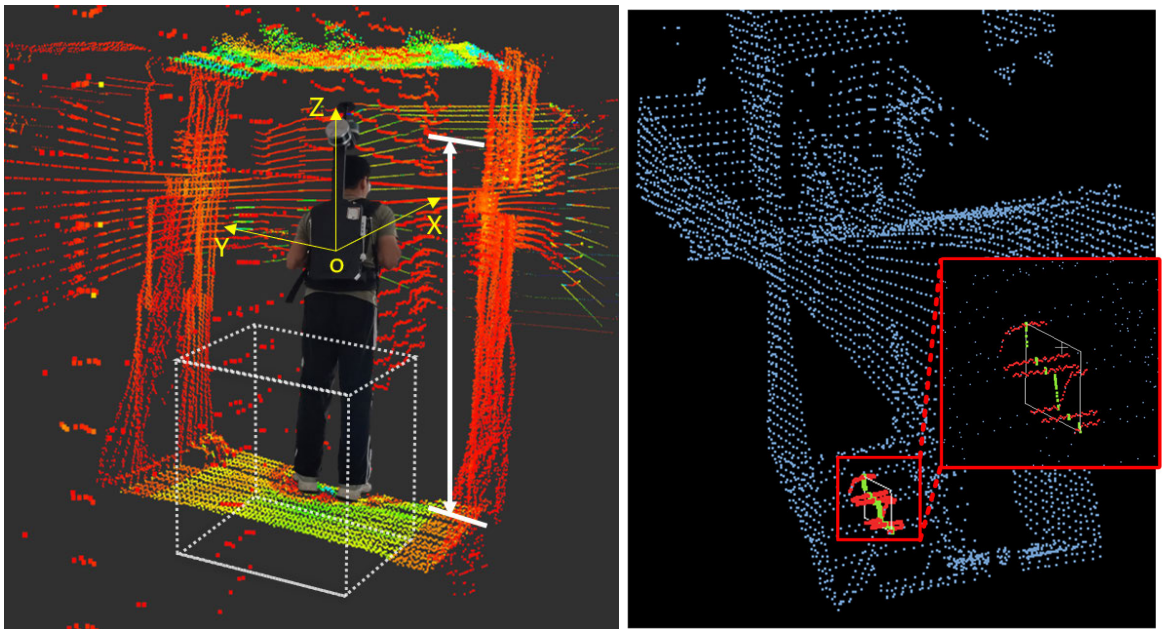
1) The motion of BLS from plane represented by π^0 to the staircases could be described as the motion from x_2 to x_3. The ground constraint from π^0 will be interrupted once the RASR recognizes the staircases or a large translation on the z-axis occurs. During walking on stairs, only binary edges provided by laser odometry will be added to the pose graph. These constraints could be represented by (12) and (13). In Fig. 6, the graph edges connecting x_2 with x_3 and the edges between x_4 with x_5 illustrate these situations

$$x_{k-1}^k = \left(x_0^{k-1} \right)^{-1} \cdot x_0^k = \left(x_0^{k-1} \right)^T \cdot x_0^k \tag{12}$$

$$e_k^o \left(x_{k-1}^k, f_o^{k-l} \right) = \sum_{j=1}^{m} \left| p_{o,j}^{k-1} - \left(R_{k-1}^k \cdot p_{o,j}^k + t_{k-1}^k \right) \right|^2. \tag{13}$$

2) The new ground constraint will be constructed after the determination of plane coefficients. In case within Fig. 6, the BLS moves from graph node x_5 and x_6. When a flat ground plane was detected by RASR after descending from stairs, the points of the ground plane were segmented with RANSAC to solve the normal vector of the plane. The solved normal vector will be aligned to [0, 0, 1] to determine the constant intercept of plane coefficient π^6. The determination of the ground plane coefficients is described in (14) and (15). The odometry edge will be also added to the pose graph

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(a) (b)

Fig. 4. Cube drawn with white dotted line in (a) shows the clip box. The yellow coordinate axis demonstrates the current scan coordinate. The tiny and great intercept of the box with the x-, y-, and z-axes of the coordinate are represented by $[x^b_{\text{cb}}, y^a_{\text{cb}}, y^b_{\text{cb}}, z^a_{\text{cb}}, z^b_{\text{cb}}]$. The red points are labeled by yellow box in (b) shows the extraction result. And the green points are the projected points.

using (12) and (13). It is important to note that the weight corresponding to d' in information matrix Ω^N_k described in (15) is fixed at zero. The constant term d' in fit ground plane coefficients $\pi' = [n_x{}', n_y{}', n_z{}', d']^T$ is determined by LO and the alignment of the normal vector instead of d' in π'. To maintain the consistency of the floor plane, the previous plane coefficients are retrieved. If the deviation between π^k and historical coefficients π^i is less than threshold value Δd_f, π^k will be assigned as π^i

$$\pi' = \left(n', d'\right)^T = (0, 0, 1, 0)^T \tag{14}$$

$$e^N_k\left(x^k_{k-1}, \pi^k\right) = \tau\left(\pi^k\right) - \tau\left(\pi'\right) \tag{15}$$

$$\Omega^N_k = \begin{bmatrix} \sigma_g^{-1} & 0 & 0 \\ 0 & \sigma_g^{-1} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{16}$$

3) After the determination of the ground plane coefficients as earlier, all pose nodes will be connected with the ground plane node in 2) before the next staircase is recognized. During the motion from x_7 to x_8, the floor points are segmented to fit the floor plane coefficient π^7 and π^8, which will be optimized to minimize the deviation with π^k as described in (17) and (18). For each graph node constructed on the flat ground, ground plane π^j was fit by RANSAC using the points from the entire field of view (FOV) in a specific height range $[z^a_{\text{gs}}, z^b_{\text{gs}}]$. The corresponding weight for d^k in information matrix Ω^G_k is set to σ_g^{-1} in (19). Subsequently, the d^k is optimized close to d^j using (18). In Fig. 6, graph nodes x_{0-2} and x_{7-8} illustrate this situation

$$\pi^k = \left(n', d'\right)^T = \left(0, 0, 1, d^k\right)^T \tag{17}$$

$$e^G_k\left(x^k_{k-1}, \pi'\right) = \tau\left(\pi^k\right) - \tau\left(\pi^j\right) \tag{18}$$

$$\Omega^G_k = \begin{bmatrix} \sigma_g^{-1} & 0 & 0 \\ 0 & \sigma_g^{-1} & 0 \\ 0 & 0 & \sigma_g^{-1} \end{bmatrix}. \tag{19}$$

G. Loop Detection for Preventing Wrong Matching

In the map built by LO, there may be drifts due to LiDAR measurement noise and algorithmic error, especially in nonstructured environments, such as multistory buildings. To correct for these drifts and improve mapping accuracy, loop closure detection is necessary. However, the similar structure between adjacent and nonintervisible stories easily leads to a wrong loop matching [see Fig. 7(a) and (b)]. Benefiting from the low-drift positioning estimated by FGO with MGCPO, we adjust the search range in the z-axis direction to prevent incorrect loop matching [see Fig. 7(c) and (d)]. The loop detection is also implemented based on FAST-GICP registration, as described in (20). Initially, a set of historical keyframes are selected using a sphere with radius r_l which is centered on positioning initial x_k provided by FGO with MGCPO [see Fig. 7(c)]. Historical keyframes with a position deviation in the z-axis direction greater than r_z from x_k are excluded from consideration. The current keyframe in odometry coordinate f^k_o is aligned with the candidate keyframe on odometry coordinate f^j_o. From (21) and (22), fast library for approximate nearest neighbors (FLANN) $(p^k_{o,i}, f^j_k)$ denotes that we use the FLANN search algorithm [35] to retrieve the transformed frame f^j_k to find the nearest neighbor of point $p^k_{o,i}$

$$x^k_j = \underset{x^k_j}{\text{argmin}} \sum_{i=1}^{n} d_i^T \left(C_i^{f^j_o} + x^k_j C_i^{f^k_o} x^{k\,T}_j\right)^{-1} d_i \tag{20}$$

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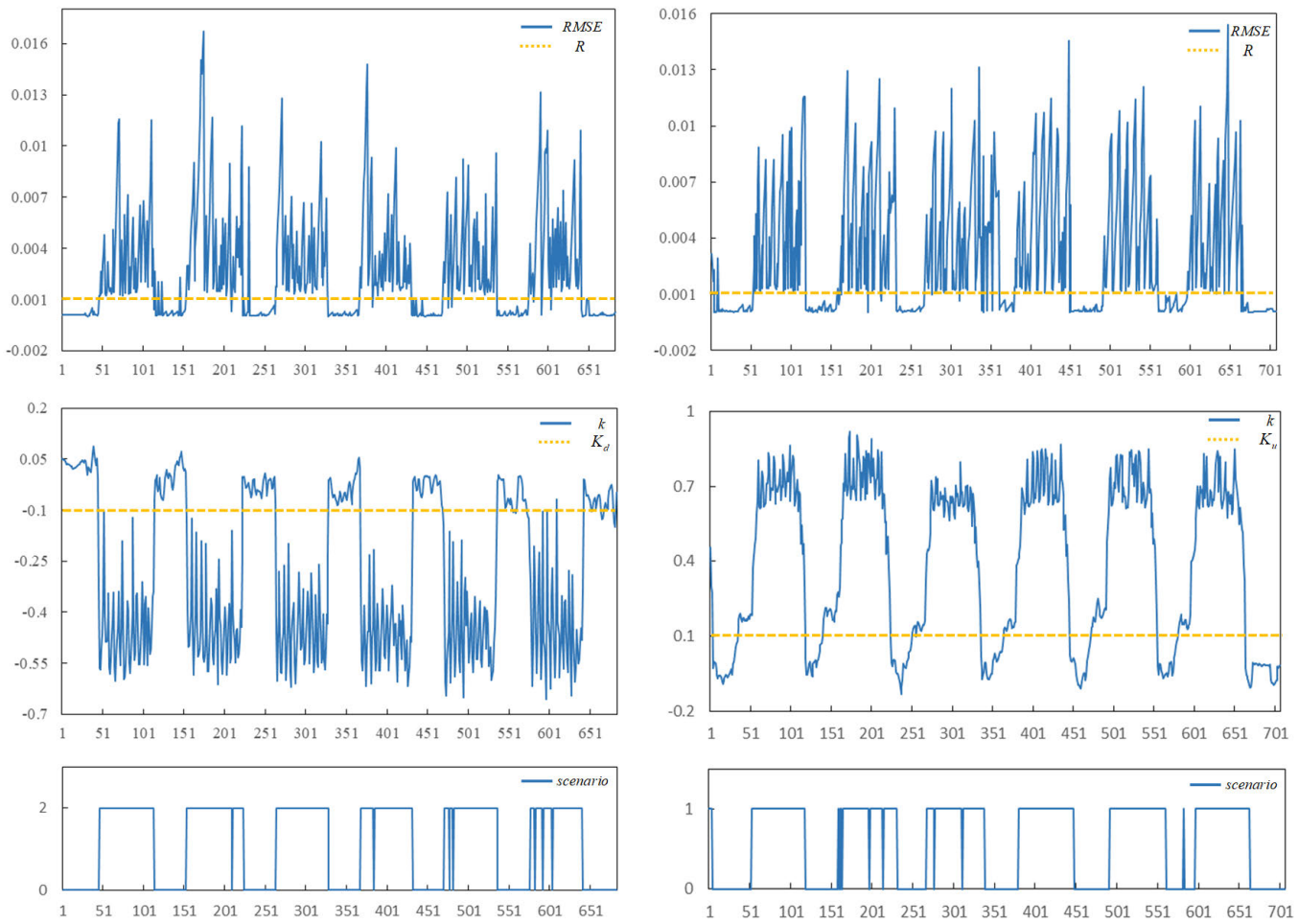


Fig. 5. Top to bottom, there are diagrams of RMSE, k and the recognition result, respectively. The diagram in left bottom is the result of descending stairs, and right is ascending stairs. The value zero indicates the current passing scenario is a flat ground; otherwise, it is a staircase.

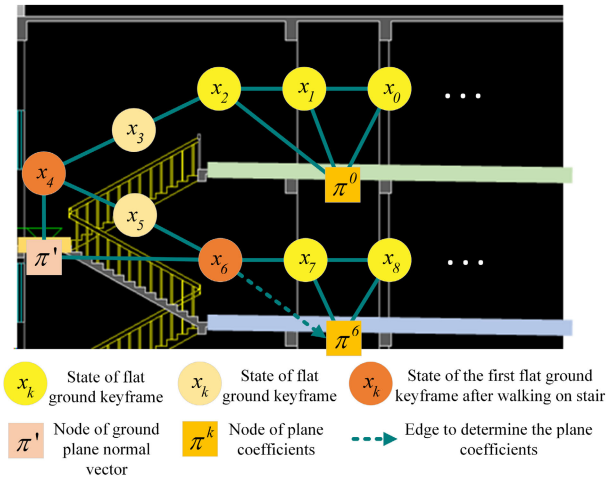


Fig. 6. Multiple ground constraints-based pose optimization strategy in cases of descending stairs.

$$f_k^j = x_j^k \cdot f_o^j \tag{21}$$

$$s_l = \frac{1}{n}\sum_{i=1}^{n}\left| p_{o,i}^k - \mathrm{FLANN}\left(p_{o,i}^k, f_k^j\right)\right|. \tag{22}$$

Finally, the loop matching frame is determined based on the fitness score calculated by registration. If the lowest fitness score is smaller than the threshold, the corresponding historical keyframe f_o^j is considered as the loop matching frame. Subsequently, an edge for loop correction will be added to the pose graph as described in the following:

$$e_k^L(\Delta x) = \sum_{i=1}^{m}\left| p_{o,i}^j - \left(R_0^k \cdot p_{o,i}^k + t_0^k\right)\right|^2. \tag{23}$$

After adding a loop closure edge constraint into the pose graph, the joint residual function is constructed. Subsequently, a global graph optimization is performed to obtain a globally consistent trajectory and map. Where Δx denotes the relative transformation sets between each pose node, $p_{o,i}^j$ denotes point i within f_o^j, R_0^k denotes the rotation of pose node x_k, and t_0^k is corresponding translation.

III. Experiments

In this section, we first present the experimental setup and parameter settings. Then, we conducted two groups of experiments to evaluate the robustness of the RASR method. We also performed two additional sets of experiments to analyze the SLAM performance of the proposed method. All

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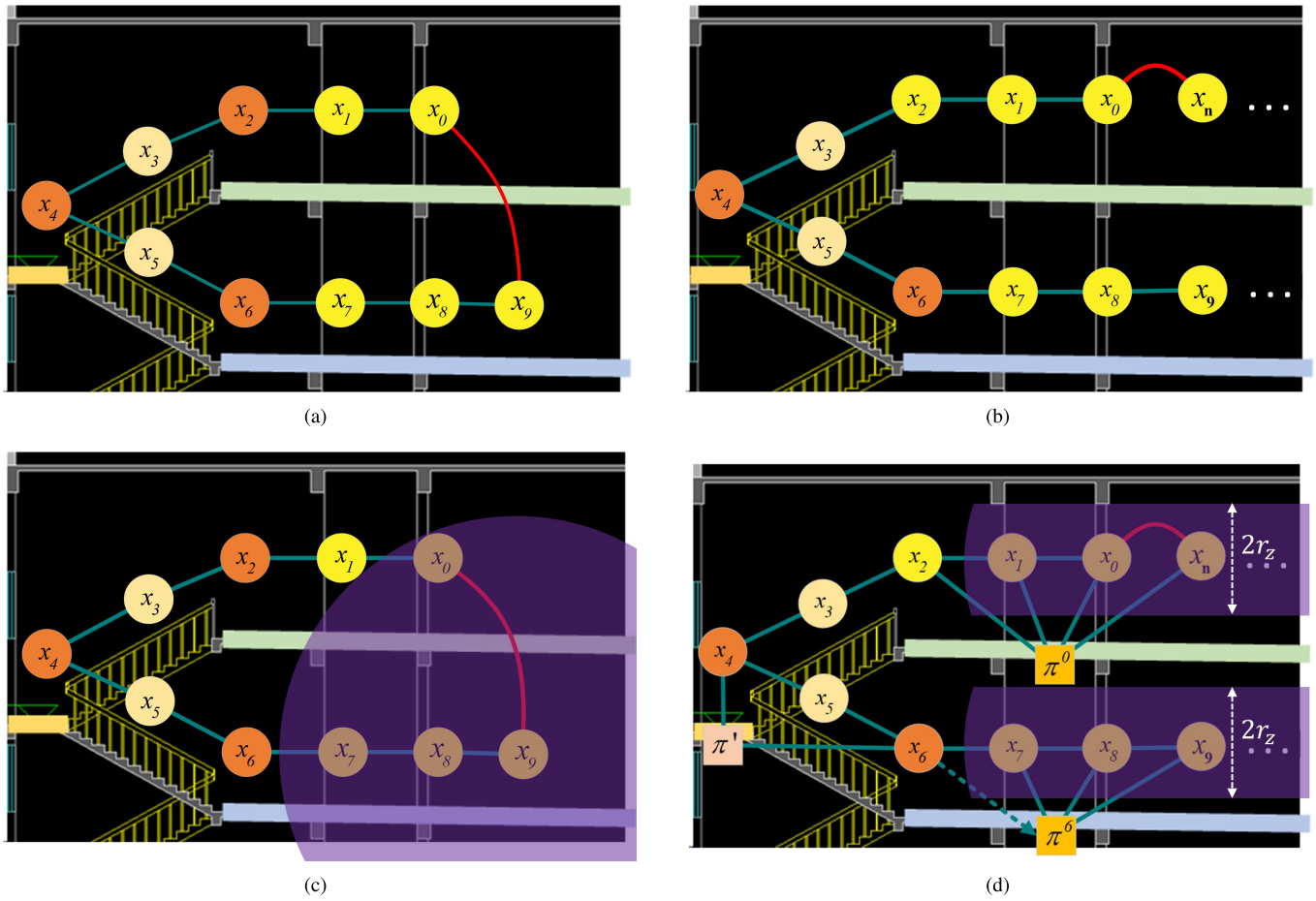


Fig. 7. (a) and (b) Demonstrate the pose graph of incorrect and correct loop correction in similar texture environments, respectively. In (c), the purple circle represents the search range for loop matching. The adjustment of the circle's radius is unreliable because it is hard to balance the drift of LO and limitation between two different stories. Then, an incorrect might occur. In (d), the LO drifts in the z-axis direction are compressed so that the search range in the z-axis direction could be reduced. Then, an incorrect loop closure could be avoided.

datasets used in the experiments are collected by our backpack LiDAR system (BLS). We compare the mapping performance of our approach with three state-of-the-art methods and provide a comprehensive analysis of the MGCPO. Additionally, we compare some sample points in the map generated by the proposed method to the ground truth obtained using a ruler and prior architectural knowledge. All experiments are done on an Intel Xeon E5-2690 2.60-GHz computer with 28 cores. Table I shows general information about captured datasets.

A. Configuration

In the data preprocessing module, a voxel filter with a voxel size of 0.4 m is used to downsample a new coming scan. The parameter L_s is set to 12 to maintain the latest 12 keyframes for scan-to-submap registration. In the LO module, the parameters for submap maintenance $[\omega, D_t]$ are set as [2.0, 0.25] (rad/m) (Section II-D). In the scenarios recognition module, the clip box setting $[(x^a_{\text{cb}}, x^b_{\text{cb}}), (y^a_{\text{cb}}, y^b_{\text{cb}}), (z^a_{\text{cb}}, z^b_{\text{cb}})]$ and the recognition parameters $[(K_d, K_u), R]$ have been discussed in Section II. The clip box $[z^a_{\text{gs}}, z^b_{\text{gs}}]$ is set as $[-2.2, -1.8]$ m to extract the ground point cloud for fitting planar coefficients by RANSAC. In the pose optimization module, to balance optimization frequency and maintain the sparsity of the pose graph, the graph update interval, denoted as T_G is set to 0.2 s. Similar to submap management in scan matching, if the rotation or displacement of the current frame with respect to the previous frame is larger than the threshold's value [2.0, 0.5] (rad/m). We set the threshold for Δz (Section II-F) as ± 0.13 m, which corresponds to the height of a stair step in our experimental scenarios. After merging the point clouds, 32-beam scans were input to A-LOAM, LeGO-LOAM, M-LOAM, and our method. For compatibility, our datasets are transformed to appear as 32-beam single-LiDAR datasets by executing the corresponding ROS launch files.

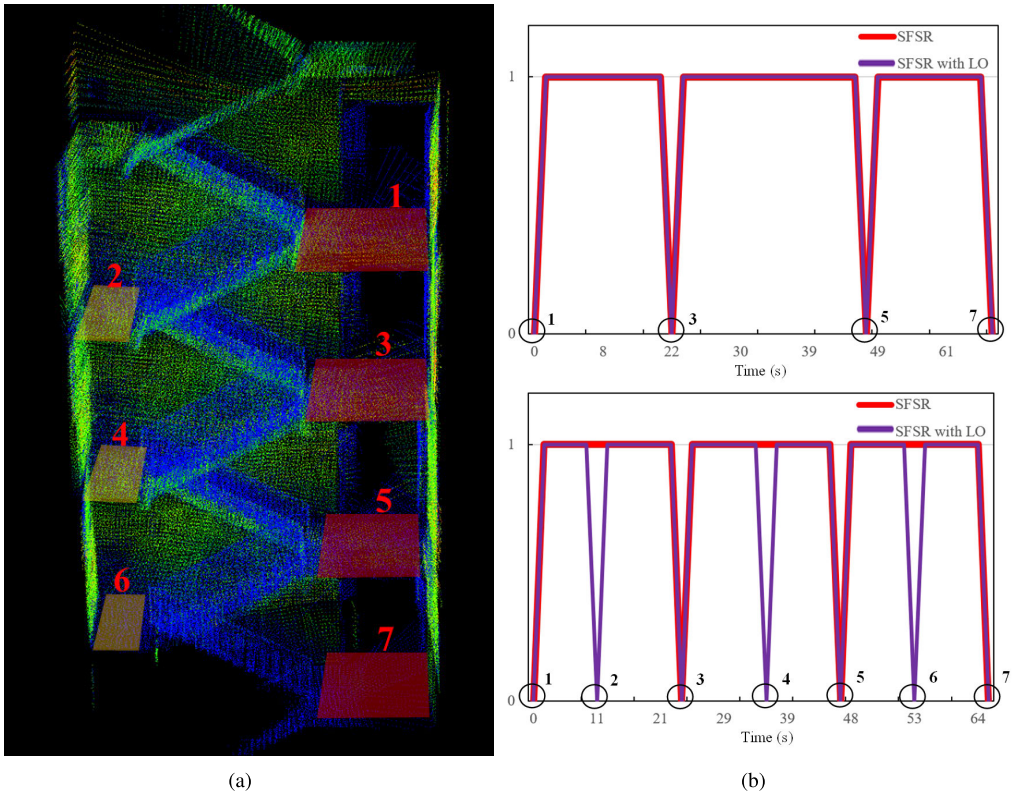
B. Accuracy Evaluation for Scenarios Recognition

In this section, we perform two groups of experiments to demonstrate the performance of scenario recognition. The first group of experiments utilizes datasets 01, 02, and 03 to analyze the recognition accuracy of RASR in our framework. Dataset 01 consists of the logs from a short corridor and a turning corner that contains a flat ground. Datasets 02 and 03 correspond to a short staircase environment with walking upward for dataset 02 and walking downward for dataset 03. In an indoor environment, usually, the number of steps in a flight of stairs is typically limited. Therefore, the stair mileage

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TABLE I
GENERAL INFORMATION ABOUT CAPTURED DATASETS

Type	ID	Key characteristics	Number of Scans	Travel Distance (m)
Unique scene	01	Flat ground in a narrow corridor	263	16.8
	02	Walking upward on staircase	68	3.7
	03	Walking downward on staircase	77	3.3
Mixed scene	04	Walking upward on staircase from 12th to 15th floor	942	37.9
	05	Walking downward on staircase from 15th to 12th floor	967	37.9
Large scale multi-storey scene	06	Three adjacent floors with similar textures, the path contains more than 2 loop points	4831	290.6
	07	Three discontinuous floors with similar textures, the path containing more than 2 loop points	7446	346.8
Indoor and outdoor mixed environment	08	Intervisible two storey environments including a loop point	2048	149.2
	09	Indoor and outdoor mixed environment including a non-intervisible corridor	3430	205.0



(a) (b)

Fig. 8. Experimental scenarios and results of RASR. Seven target ground planes are labeled by red and yellow quadrilaterals in (a). (b) Top: Planes 1, 3, 5, and 7 are detected on dataset 04 with or without assistance of motion information. The black circle in bottom shows that planes 1, 3, 5, and 7 are detected by RASR on dataset 05. With assistance of LO, all planes are recognized.

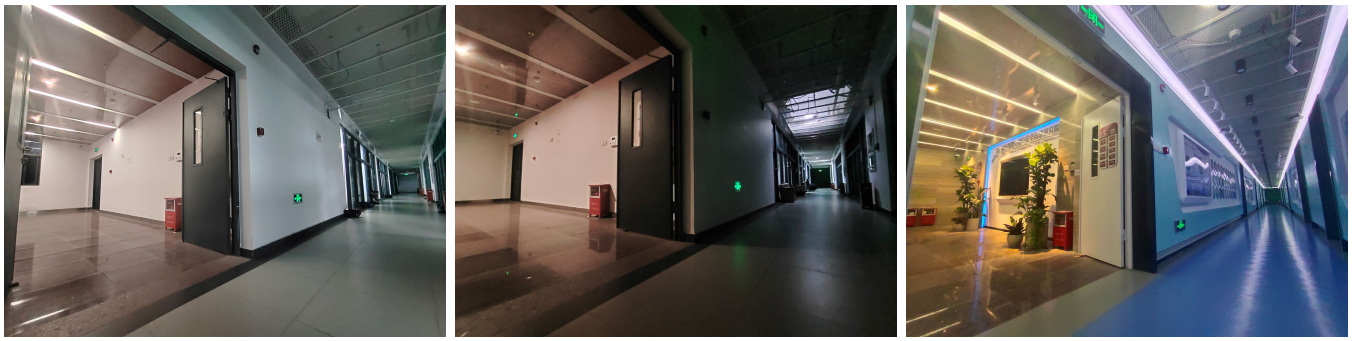
in these datasets is also limited. To reduce the randomness of the experiment and expand the sample, we ran the algorithm in datasets 01, 02, and 03 five times, respectively. Table II presents the confusion matrices calculated for datasets 01, 02, and 03, respectively. The accuracy and intersection-over-union (IoU) are defined as (24) and (25). Where TP and FN represent true positives and false negatives, respectively. Table II provides an overall summary of the scene recognition results. It can be observed from Table III, the recognition accuracy in dataset 02 is higher than that in dataset 03. This difference may be attributed to the larger ground extraction area when the operator is walking upward on the stairs

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}} \tag{24}$$

$$\text{IoU} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}}. \tag{25}$$

Based on the result of RASR, scene categories are further inferred with the assistance of motion in the z-axis direction from LO, but on a lower frequency in MGCPO. The frequency

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(a) (b) (c)

Fig. 9. Some similar textures in environments of dataset 06. Indoor corridor in (a) 15th, (b) 14th, and (c) 13th floor of the building.



(a) (b) (c)

Fig. 10. Left: trajectory generated by LOAM. Middle: trajectory generated by M-LOAM. Right: trajectory generated by FGO. (a) Different storey crossed in LOAM's trajectory. (b) Different storey crossed in M-LOAM's trajectory. (c) No incorrect cross occurred in FGO's trajectory.

is determined by the parameter of node adding, which leads to a sparser scene recognition result. The second group of experiments aims to demonstrate the scene recognition ability of the whole module in longer and mixed scenes. The amount of planes successfully recognized is considered a criterion to evaluate the recall performance of plane detection in MGCPO. As depicted in Fig. 8(a), the scene consists of a staircase and seven flat ground planes. Among them, the area of planes 2, 4, and 6 is about 3.2 $\times$ 1.5 square meters, and that of planes 1, 3, 5, and 7 is 3.1 $\times$ 3.0 square meters. Datasets 04 and 05 are used to simulate upward continuous walking and downward continuous walking of operators in staircase environments. As can be seen from the Fig. 8(b), the RASR with LO successfully extracted all seven ground planes in dataset 05. RASR captures four large ground planes. In dataset 04, both RASR and RASR with LO extract four large ground planes in the route. The experiment demonstrates that LO provides assistance in improving accuracy, particularly in the downward walking scenario. Table II shows the confusion matrices for scenario recognition, in which the top row denotes the predicted types and the left column denotes the ground truth types.

TABLE II

CONFUSION MATRIX FOR SCENARIOS RECOGNITION, IN WHICH THE TOP ROW DENOTES THE PREDICTED TYPES AND THE LEFT COLUMN DENOTES THE GROUND TRUTH TYPES

dataset	1		2		3	
	plane	stair	plane	stair	plane	stair
plane	1230	0	0	0	0	0
stair	0	0	43	292	82	283

TABLE III

RESULTS OF SCENES CLASSIFICATION

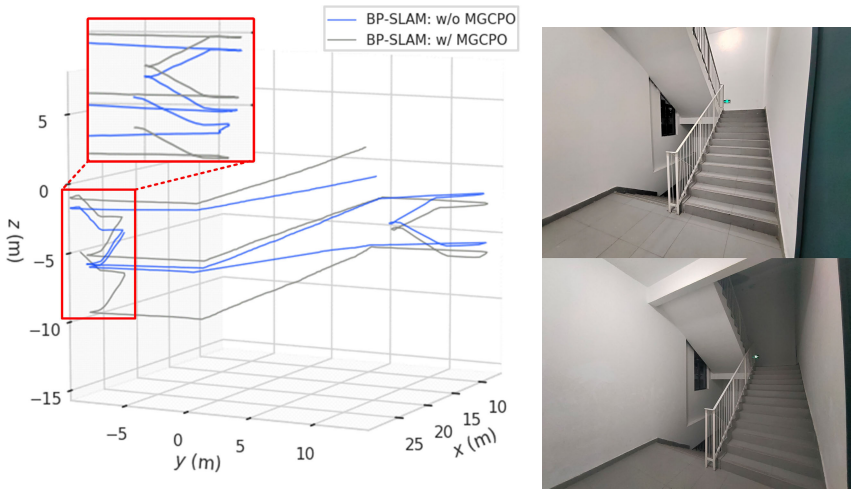
Dataset	1	2	3
Accuracy	100.00%	87.16%	77.53%
Categories	plane	stair	-
IoU	90.77%	82.14%	-

C. Analysis of SLAM Performance

In this section, we evaluate the algorithm using datasets 06 and 07 to assess the impact of MGCPO on loop detection and mapping performance. Two datasets were collected in a multistory building where each floor has a flat corridor. The operator walked in the "8" pattern and more than two loop closure points were contained in the path. Details about the datasets can be seen in Table I.

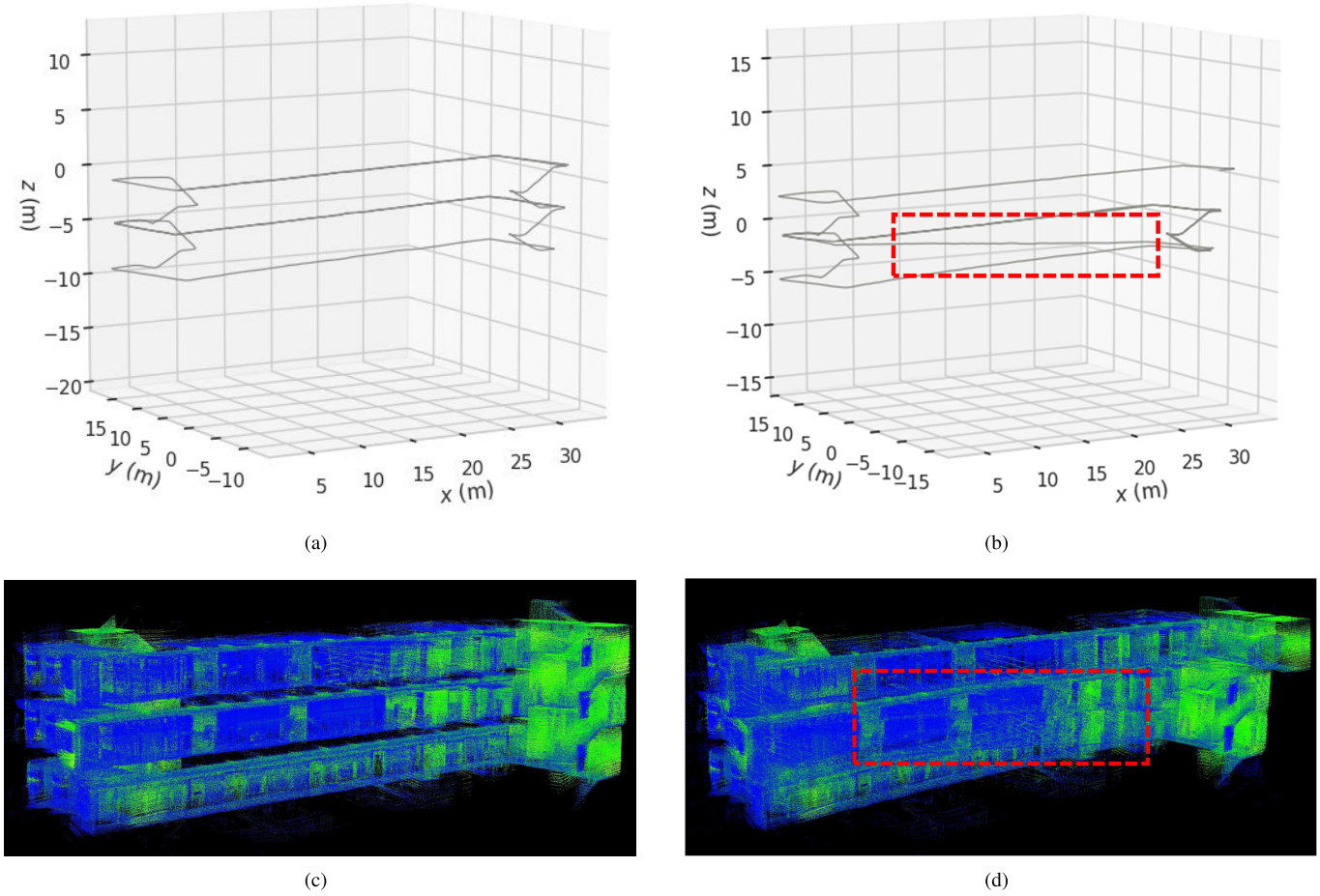
1) Effect of MGCPO on Loop Detection: The first experiment is performed on dataset 06 which represents challenging scenarios. The scanning route comprises three adjacent floors with similar textures [see Fig. 9(a)–(c)]. We first compare the trajectories generated by LOAM, M-LOAM, and FGO. LeGO-LOAM did not provide a meaningful result in the experiment. In the trajectories generated by A-LOAM and M-LOAM, the corridors of different floors intersect in the regions labeled by red boxes [see Fig. 10(a) and (b)]. These drifts of 3-DOF pose [z, roll, pitch] can potentially result in a wrong loop matching between different but nearby floors due to the similar structure environment of stories. Although FGO's path preserves the

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(a) (b)

Fig. 11. Wrong loop correction matches the staircase with 13th floor to staircase on 14th floor. In tiny dash box of (a), the blue line overlaps but the gray line remains parallel. Top in (b) is the staircase of 14th floor. And the bottom is the staircase of 13th floor.



(a) (b)

(c) (d)

Fig. 12. When MGCPO is turned ON, the relationship between the three stories is clear in (a) and (c). In (b) and (d), the trajectory contains an intersection and the skeleton of stories is not clear when MGCPO is turned OFF.

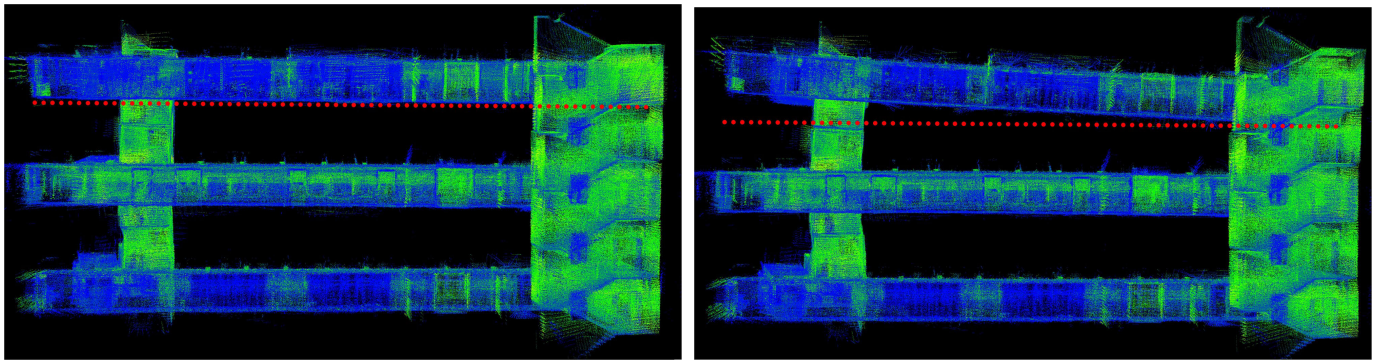
relationship between these floors, there is still noticeable drift in the z-axis direction, which makes it challenging to adjust the loop detection's search parameters.

To assess the impact of MGCPO, we varied the parameter r_z in the range of 0.3–0.8. And incorrect loop closures occurred in all six tests when the MGCPO was disabled. Fig. 11 shows a representative example of incorrect mapping results following an erroneous loop closure. It is worth noting that the enlarged block diagram in Fig. 11(a) depicts the initial positioning estimates prior to the loop closures. The incorrect

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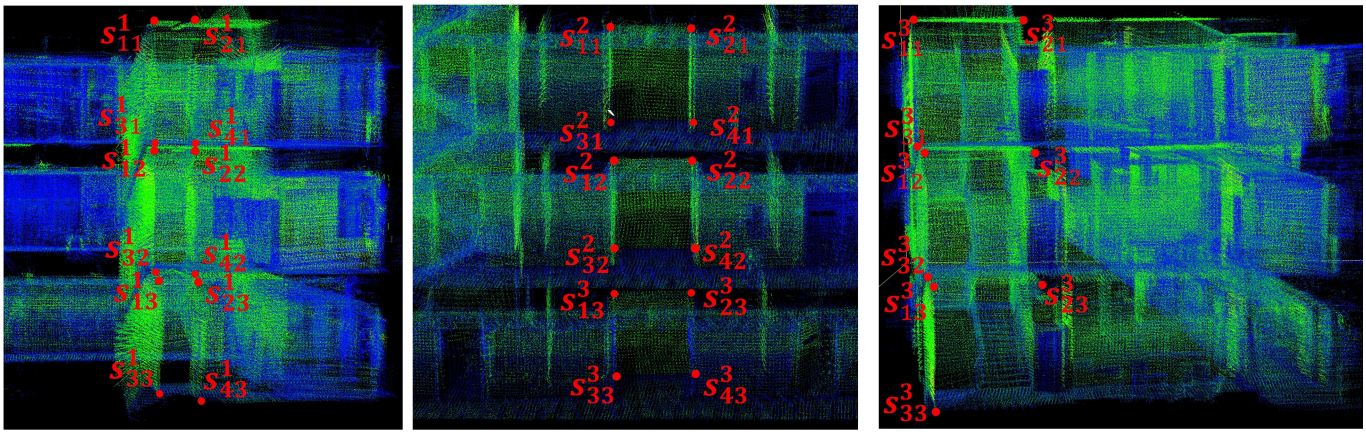
TABLE IV
TIME EFFICIENCY ANALYSIS OF OUR ALGORITHM (UNIT: s)

Method	Preprocessing	Laser odometry	Floor detection			Pose optimization			All
	filtering	Registration	clip	recognition	fitness	graph update	Loop	optimization	-
w/ MGCPO	0.0214	0.0833	0.008	0.0008	0.0034	0.0811	0.0555	0.022	0.2755
			0.0121			0.1586			
w/o MGCPO	0.022	0.0812	0	0	0	0.0809	0.2481	0.024	0.4562
			0			0.353			



(a) (b)

Fig. 13. Results of the proposed method with or without MGCPO using dataset 07. The red dashed line represents the horizontal plane. In (a), the MGCPO is turned ON. In (b), that is turned OFF.



(a) (b) (c)

Fig. 14. Three groups of selected control points for mapping evaluation. (a) Corner points group in staircase rooms of diverse floors. (b) Corner points group in tiny rooms of diverse floors. (c) Corner points group in another staircase rooms of diverse floors.

loop closures were attributed to the z-axis drift and the presence of similar textures in the staircase. Fig. 12(a)–(d) shows the mapping results and corresponding trajectories with or without MGCPO using dataset 06.

To analyze the time cost of MGCPO, we conducted the second experiment on dataset 07. In this dataset, the distance between nearby floors is increased to avoid wrong loop closures when MGCPO is disabled. And r_z is set as 0.5. A detailed runtime analysis is presented in Table IV. The scenario recognition module is able to operate at a speed exceeding 60 Hz. Importantly, the time cost of loop detection is significantly reduced when MGCPO is enabled. This is attributed to the early detection of high-scoring loop candidates, facilitated by a more accurate initial positioning estimate. In addition, Fig. 13 provides a visual comparison of the final mapping results comparison of BP-SLAM with or without MGCPO. By contrast, the result of BP-SLAM with MGCPO exhibits consistent ground plane alignment on the 13th floor.

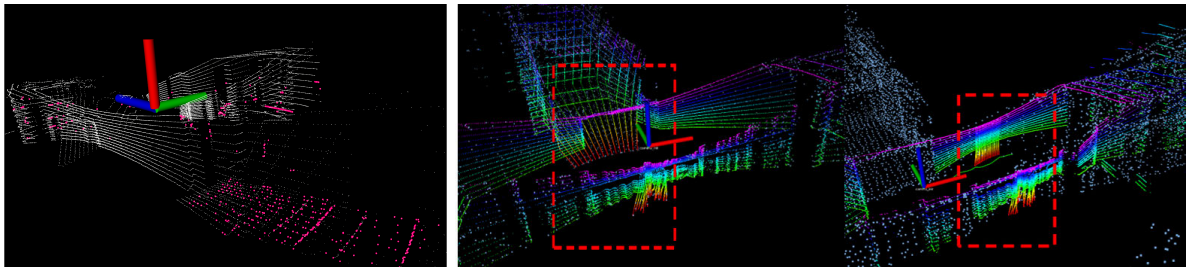
2) Quality Evaluation for Sample Points: To the best of our knowledge, there are limited open datasets for the evaluation of the multi-LiDAR SLAM method or BLS SLAM methods [36], [37]. However, these datasets are hard to evaluate the proposed methods due to the absence of ground truth, different deployment poses of laser scanners and diverse applicable environments.

Inspired by Gong et al.'s [3] work, we conducted a quantitative evaluation of mapping results in this section. We selected

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TABLE V
MAPPING ERROR ANALYSIS OF BP-SLAM

Dataset	Criterion	BP-SLAM(w/o MGCPO)				BP-SLAM(w/ MGCPO)			
		group1	group2	group3	Mean	group1	group2	group3	Mean
6	error ratio	-	-	-	-	1.0118	1.0036	1.0164	1.0106
	error rate	-	-	-	-	1.18%	0.36%	1.64%	1.06%
7	error ratio	0.9714	0.9651	0.9627	0.9664	0.9688	0.9612	0.9674	0.9658
	error rate	2.86%	3.49%	3.73%	3.36%	3.12%	3.88%	3.26%	3.42%



(a) (b)

Fig. 15. (a) Most of the points on flat ground, but some outliers on walls, are extracted by LeG0-LOAM. In (b), the comparison of laser beams in red dashed box shows the loss of the measurement in LOAM.

three groups of control points from our mapping results of datasets 06 and 07. Fig. 14 illustrates the selection process. For the control points $S^g = [s^g_{11}, s^g_{21}, \ldots, s^g_{m3}]$ in each group g, we categorized them into sets j based on the floor they belong to. Then, the Euclidean distance $d(s^g_{ij}, s^g_{ik})$ between a point s^g_{ij} and the corresponding point s^g_{ik} in another floor is calculated. Considering the architectural characteristics of multistory indoor environments, we specifically choose the corner points located at the strictly consistent decoration or staircase rooms. Thus, the corresponding ground truth $d'(s^g_{ij}, s^g_{ik})$ of the distance between points pair $d(s^g_{ij}, s^g_{ik})$ prior architectural knowledge of the layer height. To provide a quantitative error analysis, the error ratio is defined as follows:

$$r_e\left(S^g\right) = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=1}^{j-1} d\left(s^g_{ij}, s^g_{ik}\right)}{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=1}^{j-1} d'\left(s^g_{ij}, s^g_{ik}\right)} \tag{26}$$

where m is the points amount in the sets for the group. n denotes the number of stories in the experimental environments. And the error rate $\text{rate}(S^g)$ is defined in the following:

$$\text{rate}\left(S^g\right) = \left|1 - r_e\left(S^g\right)\right|. \tag{27}$$

The quantitative analysis result is shown in Table V. It is evident that the error in dataset 06 is significantly lower than that in dataset 07. Furthermore, when the MGCPO is disabled, BP-SLAM fails to generate a meaningful result due to incorrect loop correction on dataset 06. This can be attributed to the shorter distance between adjacent corridors in dataset 06 compared with dataset 07. The compact flat ground planar in the corridors can provide high-quality ground constraints, effectively limiting drift in [roll, pitch, z] through the dense MGCPO. In the case of dataset 07, the distance between two corridors with flat ground is increased, resulting in a sparser MGCPO. As a result, similar precision is achieved when executing the proposed method with or without MGCPO in dataset 07. However, MGCPO ensures that the corridors in different stories are parallel to each other (see Fig. 13).

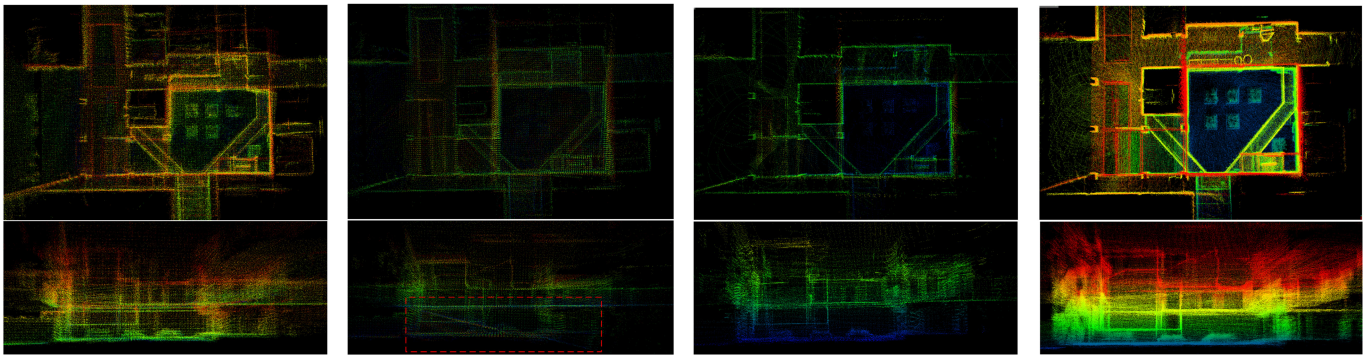
IV. DISCUSSION

To further investigate the relative strengths and weaknesses of various approaches, we conducted two experiments in an indoor and outdoor mixed environment constituted of two intervisible floors. We properly adjust the configuration of the selected methods with a pose graph implemented for a comprehensive discussion. We disguised BLS datasets as 32-beam LiDAR datasets to input the approaches. It is notable that we maintained the configuration of ground extraction for 16-beam LiDAR in SC-LeGO-LOAM because we hope to reserve the author's devise as far as possible meanwhile measurements were lost [see Fig. 15(a)].

In the first experiment, SC-A-LOAM, the advanced implementation of LOAM with loop correction, produced a consistent result. SC-LeGO-LOAM failed to estimate the vertical pose variation on outdoor stairs and generated a wrong result [see Fig. 16(f)]. We inferred the reason is that the image projection-based feature filter of LeGO-LOAM further decreases the surface features which are able to provide vertical motion observablity. M-LOAM avoided the measurement loss and produced an accurate trajectory and map due to the design of primary LiDAR and auxiliary LiDAR and uncertainty estimation of measurements. A result similar to SC-A-LOAM's is worked out by our method with MGCPO (see Fig. 16). By comparison, the thinner skeleton of the walls and footbridge demonstrates M-LOAM's outstanding pose estimate accuracy on yaw angle. In the intervisible scene, the local map accumulated from the origin in M-LOAM is more advantageous for registration than the sliding-window strategy adopted by our work.

In the second experiment, a section of nonintervisible corridor was added to the path [see Fig. 17]. SC-LeGO-LOAM

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(a) (b) (c) (d)

Fig. 16. (a)–(d) Top view and side view of mapping results produced by SC-A-LOAM, SC-LeGO-LOAM, M-LOAM, and BP-SLAM, respectively.

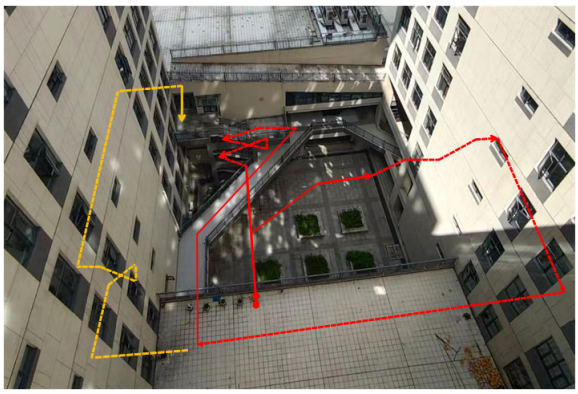
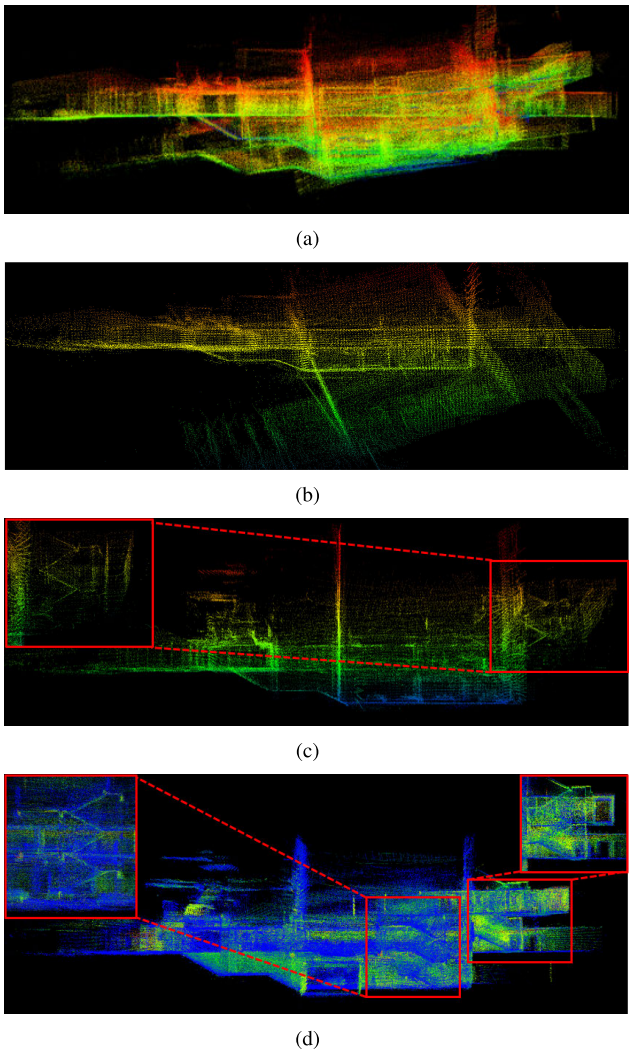


Fig. 17. Red line and arrow represent the trajectory in the first experiment. The indoor corridor is noted by the orange arrow. The red point represents the starting and ending of the trajectory.

drift away in the corridor did not generate a meaningful result [Fig. 18(b)]. SC-A-LOAM detected the loop but failed to optimize the pose correctly [see Fig. 18(a)]. The area labeled by the red box in Fig. 18(c) shows that a drift on the z-axis direction occurred in the nonintervisible corridor despite consistent ground on origin generated after pose graph optimization. The consistent ground plane and horizontal corridor show that a low-drift SLAM result is obtained by BP-SLAM with MGCPO [Fig. 18(d)].

The vertical drift usually occurs in the nonintervisible and narrow indoor environments though more measurements were inputted to these feature-based methods. However, these methods maintained a robust performance to estimate yaw angle. In our opinion, it might be caused by two factors as follows. The features were usually extracted by calculating the curvature of points within each laser beam in feature-based methods. When the laser beam is horizontal, the extraction is robust in wide scenes that contain sufficient vertical edge features, such as lampposts, the surface of outdoor buildings, boundary, and the surface of wall indoor. These features offer reliable observability to estimate yaw angle and translation on the x- and y-axes. However, the estimate of another 3-DOF pose [roll, pitch, z] usually relies on the planar points from the ground. In our BLS case, horizontal LiDAR could only capture planes from ceilings and floors in indoor scenes which are nonconsecutive in narrow staircases and corners. These



(a)

(b)

(c)

(d)

Fig. 18. (a)–(d)Side view of SC-A-LOAM, SC-LeGO-LOAM, M-LOAM, and BP-SLAM's mapping results. The area labeled by red box in (c) shows that M-LOAM drifted in the indoor corridor after ascending the stairs. In (d), the clear skeleton of staircases and corridors shows that BP-SLAM's result maintained consistency.

measurements could not provide a stable observability to estimate the roll angle, pitch angle, and translation on the z-axis. Second, the ring parameters of feature extraction in LOAM series might lead to a measurement loss in multi-LiDAR systems, especially with a large pitch or roll angle

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between different LiDAR [see Fig. 15(b)]. Feature-based methods might be more suitable for first utilizing observations from diverse LiDAR, respectively, for state estimate and then fusing the estimates, rather than fusing observations from various LiDAR for total state estimation, while developing multi-LiDAR systems.

Even so, in indoor scenes with elevation changes, SLAM localization might drift due to insufficient observability of motion, especially on the z-axis. The introduction of ground constraints is one of the solutions. However, the ground constraints in our work require the existence of the flat grounds which also limit the application of our work in some indoor environments with complex terrain. In these environments, local ground constraints and gravity constraints from inertial measurement units (IMUs) might work better.

V. Conclusion

This article presents a novel SLAM method for dual-LiDAR BLS, specifically designed for mapping multistory indoor environments with high accuracy. The proposed method utilizes a fixable pose optimization strategy, incorporating multiple ground plane constraints into the pose graph based on the current scenario. A geometric feature-based method for dynamic scene classification, which accurately distinguishes three different types of walking behaviors: walking upstairs, walking downstairs, and walking on a flat surface is proposed. The scene classification results are then applied as constraints for localization and loop closure detection in LiDAR-based SLAM processes. By leveraging this strategy, incorrect loop closure can be avoided and the algorithm ensures better consistency among floor planes in multistory environments. The experimental evaluation was conducted using two different datasets that represent typical multistory indoor environments. To ensure a fair comparison, state-of-the-art methods were adjusted to utilize the datasets collected with our BLS and their performance was compared with the LO in our proposed method. The results demonstrate that the proposed method achieves accurate mapping in multistory indoor environments, leveraging the capabilities of our BLS.

Moreover, the method proposed in this article has been mainly validated in indoor and semiindoor/semioutdoor spaces. The verification of the method in outdoor environments and the refinement and in-depth discussion of its optimization still require further elaboration. In future works, we aim to expand the range of applicable scenarios by integrating IMU data to enhance scene recognition, enabling the mapping of additional environments, such as parking lots with slopes and stairs. Furthermore, inspired by [7], [8], and [31], a pose graph optimization using plane measurements and accumulations of BLS scans is also our research vision.

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Baoding Zhou (Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2015.

He is currently an Associate Professor with the College of Civil and Transportation Engineering, Shenzhen University, Shenzhen, China. His research interests include indoor localization and mapping, mobile computing, and intelligent transportation.



Haoquan Mo received the B.S. degree in traffic engineering from the Nanjing Tech University Pujiang Institute, Nanjing, China, in 2020. He is currently pursuing the M.E. degree with Shenzhen University, Shenzhen, China.

His research interests include simultaneous localization and mapping and autonomous robots.



Shengjun Tang (Member, IEEE) received the Ph.D. degree in cartography and geograph information system (GIS), from the State Key Laboratory of Mapping and Remote Sensing Information Engineering, Wuhan University, Wuhan, China, in 2017.

He held a research assistant position with The Hong Kong Polytechnic University, Hong Kong, from 2014 to 2017. He is currently an Associate Researcher with Shenzhen University, Shenzhen, China. He has long been engaged in the research of theory and applications of visual simultaneous localization and mapping, indoor 3-D mapping, and multisensor fusion modeling.



Xing Zhang (Member, IEEE) received the B.E. and Ph.D. degrees in geographic information science from Wuhan University, Wuhan, China, in 2005 and 2011, respectively.

He is currently with the Guangdong Key Laboratory of Urban Informatics, Shenzhen University, Shenzhen, China. His research interests include pedestrian navigation, indoor localization, ubiquitous computing, and mobile mapping.



Qingquan Li received the Ph.D. degree in photogrammetry and remote sensing from the Wuhan Technical University of Surveying and Mapping, Wuhan, China, in 1998.

For over 20 years, he has been engaged in the developments of surveying engineering and its innovation applications, contributing especially to the principles and practices of dynamic precision engineering surveys. His research interests include the integration of global navigation satellite systems, remote sensing (RS), and geograph information system (GIS), precision engineering surveys, and spatiotemporal big data analytics.

Dr. Li is an Academician of International Eurasian Academy of Sciences (IEAS) and a member of the Committee at the Science and Technology Commission of the Ministry of Education, China. His work has been recognized with a variety of distinctions. His research received the National Scientific and Technological Progress Second Prize (China) and the Ministry of Education (MoE) Scientific and Technological Progress First Prize (China). He is also the Chief Editor of the *Journal of Geodesy and Geoinformation Science*.

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